

A
Major Project
On
**SMART CRISIS MANAGEMENT: A UNIFIED SYSTEM FOR
EARLY WARNING, EMERGENCY SUPPLY ALLOCATION,
AND POST-DISASTER RECOVERY TRACKING**

(Submitted in partial fulfilment of the requirements for the award of Degree)

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CERTIFICATE

This is to certify that the project entitled “**SMART CRISIS MANAGEMENT: A UNIFIED SYSTEM FOR EARLY WARNING, EMERGENCY SUPPLY ALLOCATION, AND POST-DISASTER RECOVERY TRACKING**” being submitted by **T. Vaishnavi (227R1A05J8), P. Vishali (227R1A05G7) & K. Shiva Krishna (237R5A0520)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

This project is titled “Smart Crisis Management: A Unified System for Early Warning, Emergency Supply Allocation, and Post-Disaster Recovery Tracking”. Disasters, both natural and man-made, continue to pose critical threats to human life, infrastructure, and socio-economic stability. Rapid response and effective resource allocation are often hindered by fragmented communication, delayed information flow, and inadequate monitoring systems. To overcome these challenges, this work proposes a comprehensive, integrated framework for smart crisis management. The system combines real-time data collection from IoT-enabled sensors, predictive analytics, and automated decision-making to provide timely early warnings, optimize the allocation of emergency supplies, and track post-disaster recovery efforts. Advanced algorithms are used to analyze environmental, geographic, and socio-demographic data, enabling authorities to prioritize critical areas and deploy resources efficiently. A centralized dashboard allows continuous monitoring and coordination among multiple agencies, enhancing situational awareness during crises. The proposed framework was evaluated using historical disaster data and simulated emergency scenarios, demonstrating significant improvements in response time, resource optimization, and recovery tracking. Results indicate that a unified approach not only minimizes human and economic losses but also strengthens community resilience and supports data-driven decision-making in disaster management. This study highlights the potential of integrating predictive analytics, automation, and real-time monitoring to transform traditional crisis management systems into smarter, more responsive solutions.

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1. INTRODUCTION

1. INTRODUCTION

The project, titled “Smart Crisis Management: A Unified System for Early Warning, Emergency Supply Allocation, and Post-Disaster Recovery Tracking,” is designed to develop an advanced, data-driven system capable of managing disasters efficiently. The system integrates early warning mechanisms, emergency resource allocation, and post-disaster recovery tracking into a unified platform, aiming to minimize human and economic losses during crises. By leveraging real-time data collection, predictive analytics, and automated decision-making, the system ensures timely and coordinated responses to natural and man-made disasters.

Given the increasing frequency and severity of disasters worldwide, traditional disaster management methods relying on manual planning and fragmented communication are insufficient. This project offers a scalable, automated solution capable of handling large volumes of data from multiple sources, enabling authorities to respond proactively and effectively. Furthermore, the system supports continuous monitoring and decision-making during all phases of disaster management—prevention, response, and recovery.

1.1 PROJECT PURPOSE

The primary purpose of this project is to enhance disaster preparedness, response, and recovery by developing a centralized and intelligent platform for effective crisis management. Existing disaster management systems often face significant challenges such as delayed warning notifications, lack of real-time information sharing, inefficient distribution of emergency resources, and poor coordination among government agencies, emergency responders, and volunteers. These limitations can lead to increased casualties, resource wastage, and slower recovery processes. Therefore, this project aims to address these issues by providing a unified digital solution that improves communication, coordination, and decision-making during emergency situations.

The system is designed to integrate advanced technologies such as Internet of Things (IoT) sensors, predictive analytics, cloud-based data processing, and automated resource management to support proactive disaster management. By continuously collecting and analyzing real-time environmental and situational data—such as weather conditions, water levels, population density, and infrastructure status—the platform can identify potential risks and generate early warning alerts to authorities and citizens. This early detection capability enables faster evacuation planning, better preparedness, and timely response to minimize damage and protect lives.

Another important purpose of the project is to optimize the allocation and distribution of emergency supplies, including food, water, medical equipment, and rescue personnel. The system uses intelligent algorithms to prioritize affected areas based on the severity of the disaster, availability of resources, and accessibility of locations, ensuring fair and efficient resource utilization. In addition, the platform provides a centralized dashboard that allows administrators to monitor incidents, track volunteer activities, manage resources, and evaluate recovery progress in real time. This improves transparency, accountability, and coordination among all stakeholders involved in disaster management.

Furthermore, the project aims to strengthen community resilience by supporting post-disaster recovery tracking and long-term planning. By maintaining accurate records of relief distribution, infrastructure restoration, and rehabilitation efforts, the system helps authorities assess recovery performance and make informed decisions for future disaster preparedness. Overall, the purpose of this project is to create a smart, reliable, and scalable crisis management system that reduces human error, improves response efficiency, enhances collaboration among agencies, and ultimately minimizes the social and economic impact of disasters.

1.2 PROJECT FEATURES

This project incorporates several advanced and practical features designed to improve the efficiency, accuracy, and coordination of disaster management activities. These features focus on early risk detection, efficient resource utilization, continuous monitoring, and scalable system performance to support authorities, emergency responders, and communities during crisis situations.

- **Early Warning System:** The system includes an intelligent early warning mechanism that continuously collects real-time environmental, geographic, and social data from IoT-enabled sensors, satellite data, weather services, and mobile applications. Predictive analytics and machine learning models analyze this data to detect abnormal patterns such as rising water levels, extreme weather conditions, or sudden environmental changes. Once a potential threat is identified, the system automatically generates alerts and notifications to authorities, volunteers, and citizens through dashboards, SMS, or mobile applications. This proactive approach enables timely evacuation, risk mitigation, and preparedness measures, significantly reducing casualties and property damage.
- **Automated Emergency Supply Allocation:** One of the core features of the system is the automated management and distribution of emergency resources. The platform uses predictive analytics and optimization algorithms to determine the most critical areas

affected by disasters based on factors such as population density, disaster severity, accessibility of routes, and availability of resources. It then prioritizes and allocates essential supplies such as food, water, medical kits, shelter materials, and rescue equipment efficiently. This feature minimizes delays, prevents resource shortages or duplication, and ensures that relief materials reach the right locations quickly and effectively.

- **Post-Disaster Recovery Tracking:** The system provides a structured mechanism to monitor and evaluate recovery activities after a disaster. It tracks the distribution of relief materials, restoration of infrastructure, deployment of volunteers, and rehabilitation progress of affected communities. Real-time updates and performance reports help authorities assess the effectiveness of recovery operations and identify areas requiring additional support. This feature also maintains historical records of disaster response and recovery efforts, which can be used for future planning, policy-making, and continuous improvement of disaster management strategies.
- **Scalability & Real-Time Processing:** The platform is designed to handle large-scale disasters and high volumes of data from multiple sources simultaneously. By leveraging cloud-based infrastructure, distributed databases, and real-time data processing technologies, the system can support thousands of users, devices, and emergency operations without performance degradation. Advanced algorithms enable rapid data analysis, visualization, and reporting, allowing decision-makers to respond quickly to changing disaster conditions. This scalability ensures that the system remains reliable, flexible, and efficient even during large or complex emergency situations.

By integrating these features, into a single unified platform, the proposed system delivers a comprehensive, intelligent, and scalable solution for modern disaster management. It enhances early detection, improves coordination among stakeholders, optimizes resource allocation, and supports faster recovery, ultimately reducing risks, saving lives, and strengthening community resilience.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

Technological solutions for improving disaster management through better stakeholder coordination, risk identification, and efficient risk management have gained considerable attention in recent years. With the rapid growth of population, urbanization, and climate change, the frequency and severity of disasters have increased significantly. As a result, governments and organizations are focusing on advanced technologies that can help predict disasters, coordinate response operations, and support recovery processes. Modern disaster management systems increasingly rely on technologies such as artificial intelligence, the Internet of Things (IoT), big data analytics, cloud computing, and multi-agent systems to improve preparedness and response capabilities.

One important research direction is the use of predictive analytics for disaster preparedness. Marzi et al. [1] proposed a statistically based predictive model designed to assess humanitarian crisis risks by analyzing demographic trends and climate outlook indicators. Their research demonstrates how predictive models can identify regions that are vulnerable to disasters and help authorities take preventive measures before crises occur. By combining climate data, population statistics, and historical disaster records, their model supports proactive decision-making and improves disaster preparedness strategies.

Artificial intelligence has emerged as a powerful technology capable of supporting multiple phases of disaster management. Sun et al. [2] conducted an extensive literature review on the use of AI in disaster management systems and highlighted its applications in disaster prediction, emergency response coordination, and post-disaster recovery operations. AI-based systems can analyze large volumes of heterogeneous data in real time, enabling authorities to detect potential disasters early and make faster decisions. Despite these advantages, the authors note that many AI-based disaster management solutions focus only on specific stages of disaster response rather than providing a unified framework that covers the entire disaster lifecycle.

Recent research has also explored the integration of deep learning and intelligent agent technologies for disaster management. Deep learning techniques can process large datasets from various sources such as satellite images, sensor networks, and social media feeds. Hemamalini et al. [10] investigated the use of deep reinforcement learning to improve interoperability among heterogeneous disaster management components. Their study demonstrated that reinforcement learning algorithms can coordinate different system components and optimize decision-making during disaster scenarios. By enabling intelligent

coordination among different modules, the proposed approach improves system efficiency and enhances collaboration among stakeholders involved in disaster response.

Humanitarian logistics is another important aspect of disaster management that has received considerable research attention. Efficient allocation and distribution of emergency supplies such as food, water, and medical resources are critical during disaster response operations. Several decision-making strategies have been proposed to improve logistics planning in crisis situations [13]. These approaches aim to optimize transportation routes, prioritize affected regions, and ensure that relief supplies reach victims as quickly as possible.

In addition, Devadasu et al. [11] proposed a multi-agent learning model designed to support adaptive crisis management in urban environments. Their model uses intelligent agents that cooperate with each other to manage dynamic disaster scenarios. These agents can monitor environmental conditions, analyze emergency data, and assist decision-makers in selecting appropriate response strategies. Multi-agent systems are particularly useful in disaster management because they can simulate the behavior of different stakeholders such as rescue teams, emergency responders, and government agencies.

Beyond technological advancements, organizational resilience and governance policies also play a crucial role in disaster management. Adini et al. [3] emphasized the importance of resilience-based policies and frameworks that improve preparedness and response capabilities. They argue that disaster management strategies should focus not only on technological solutions but also on strengthening organizational resilience and improving coordination among institutions. Similarly, Albris et al. [6] highlighted the importance of governance structures and institutional collaboration in managing disaster risks effectively. Their research suggests that better coordination among government agencies, emergency services, and community organizations can significantly improve disaster response outcomes. However, these studies mainly focus on policy frameworks and provide limited technical implementation details.

The Internet of Things (IoT) has also become an important technology for disaster monitoring and early warning systems. IoT-based systems use networks of sensors to collect real-time environmental data such as temperature, humidity, seismic activity, and water levels. Ray et al. [4] demonstrated the potential of IoT technologies in disaster management by developing an early warning system that monitors environmental conditions and alerts authorities about potential disasters. Such systems improve situational awareness and enable authorities to respond quickly to emerging threats.

Similarly, Poslad et al. [7] introduced a semantic IoT-based early warning system that enhances contextual awareness during environmental crises. Their system integrates data from

various sensors and uses semantic technologies to interpret environmental conditions more accurately. This approach improves the reliability of disaster detection systems and supports better decision-making during emergency situations. However, these IoT-based systems mainly focus on monitoring and early warning capabilities and do not fully address issues related to resource allocation and post-disaster recovery management.

In addition to monitoring systems, optimization-based models have been proposed to improve emergency logistics and resource allocation during disaster response operations. These models aim to determine the optimal distribution of resources such as rescue teams, emergency supplies, and medical services. Several studies have explored mathematical optimization techniques to minimize response time and maximize the efficiency of disaster relief operations [12], [16].

Recent technological trends have also incorporated edge computing and cloud computing infrastructures to address scalability and latency challenges in disaster management systems. Edge computing enables data processing closer to the source of data generation, reducing communication delays and enabling real-time decision-making. At the same time, cloud computing provides scalable storage and processing capabilities for handling large volumes of disaster-related data. Hybrid architectures that combine edge and cloud computing are increasingly being used to support critical decision-making processes during disaster situations.

Aboualola et al. [5] investigated edge-enabled disaster management systems that integrate artificial intelligence with social media analytics. Their research highlights the importance of analyzing information shared on social media platforms during emergencies. Social media data can provide valuable insights about disaster conditions, affected areas, and urgent needs of victims. By combining edge computing with AI-based data analytics, the proposed system enables faster identification of crisis events and supports timely decision-making.

Another important aspect of disaster management is the involvement of human decision-makers in automated systems. Geihs [8] emphasized the concept of human-in-the-loop systems, which integrate human expertise with automated decision-making technologies. According to this approach, automated disaster management systems should support human decision-makers rather than replace them completely. By involving experts in the decision-making process, these systems can improve accuracy and ensure that critical contextual information is not overlooked.

In addition to AI and IoT technologies, remote sensing and satellite imaging have also been widely used in disaster assessment and recovery monitoring. Ou et al. [9] explored the use

of diffusion models and large language models for post-disaster remote sensing analysis. Their approach improves the accuracy of damage assessment and helps authorities evaluate the impact of disasters more efficiently. Accurate damage assessment is essential for planning recovery operations and allocating relief resources effectively.

Another emerging technology in disaster management involves the use of multi-sensor modalities and advanced positioning systems. These technologies enable the integration of data from multiple sensors such as drones, satellites, and ground-based monitoring devices. Such systems enhance situational awareness and enable better coordination among emergency responders [15]. By combining heterogeneous data sources, disaster management systems can provide a comprehensive understanding of evolving crisis situations and support more informed operational decisions.

Despite the significant progress achieved in disaster management technologies, many existing solutions still focus on isolated aspects of the disaster management lifecycle. For example, some systems focus only on disaster prediction, while others address resource allocation or recovery monitoring. The lack of integration among these systems limits their ability to provide comprehensive support throughout the entire disaster management process.

In many cases, disaster management platforms fail to ensure smooth transitions between different phases such as prevention, mitigation, response, and recovery. This fragmentation reduces the effectiveness of disaster response operations and prevents authorities from obtaining a complete view of crisis situations.

Although technologies such as artificial intelligence, IoT, optimization algorithms, and distributed computing infrastructures have improved disaster management capabilities, there is still a lack of integrated platforms that combine all these features into a unified system. Most existing solutions address only specific operational aspects and do not provide a comprehensive framework for managing disasters across the entire lifecycle.

To address this limitation, the proposed research introduces a Smart Crisis Management System, which aims to integrate early warning mechanisms, automated resource allocation, and post-disaster recovery monitoring into a single unified platform. The proposed system leverages advanced technologies such as artificial intelligence, IoT-based sensing, and data analytics to support evidence-based decision-making for disaster management stakeholders.

By providing an integrated decision support system, the Smart Crisis Management platform aims to improve disaster preparedness, enhance coordination among emergency responders, and ensure efficient recovery monitoring. This unified approach enables

stakeholders to manage disasters more effectively and supports the development of resilient and sustainable disaster management infrastructures.

2.1 REVIEW OF RELATED WORK

The management of disasters has been an important research area for many years. With the advancement of technology, several approaches have been developed to improve disaster prediction, response coordination, and recovery monitoring. These approaches range from traditional disaster management methods to modern technology-based systems using artificial intelligence, Internet of Things (IoT), and cloud computing. This section reviews both traditional and modern approaches that have been proposed in the field of disaster management.

1. Traditional Disaster Management Approaches

In the early stages of disaster management research, most systems relied on traditional methods such as manual reporting, centralized decision-making, and human-based coordination among emergency response teams. Government agencies and humanitarian organizations depended heavily on field reports, telephone communication, and manual data collection to assess disaster situations.

Although these traditional approaches helped in organizing disaster response activities, they suffered from several limitations. The process of collecting and analyzing information was slow and often resulted in delayed response during emergency situations. Additionally, traditional disaster management systems lacked real-time monitoring capabilities and were unable to effectively coordinate multiple stakeholders involved in disaster response operations.

2. Artificial Intelligence-Based Disaster Prediction Systems

Artificial intelligence has been widely used to improve disaster prediction and risk assessment. Marzi et al. proposed a predictive model that evaluates humanitarian crisis risks by analysing demographic indicators and climate outlook data. Their research demonstrates that predictive analytics can help identify regions that are vulnerable to disasters and support early preparedness planning. Such AI-based predictive models enable authorities to take preventive actions before disasters occur. However, these systems mainly focus on disaster prediction and do not address the challenges related to emergency resource allocation or post-disaster recovery management.

3. AI Applications in Disaster Response

Sun et al. presented a comprehensive review of artificial intelligence applications in disaster management. Their research highlighted the use of machine learning algorithms for

disaster prediction, emergency response coordination, and recovery planning. AI-based systems can analyze large volumes of data from different sources such as sensors, satellite images, and social media.

These systems assist decision-makers by providing timely insights during emergency situations. However, most AI-based disaster management systems are designed to address specific phases of the disaster management process rather than providing a unified framework that integrates all phases.

4. Deep Learning and Reinforcement Learning Techniques

Deep learning techniques have also been explored to enhance disaster management systems. Hemamalini et al. investigated the use of deep reinforcement learning to enable interoperability among heterogeneous components in disaster management scenarios. Their approach allows systems to learn optimal decision-making strategies in dynamic disaster environments. By improving coordination between different components of the system, reinforcement learning techniques can increase the overall efficiency of disaster management platforms. However, these approaches require large datasets and significant computational resources, which can limit their real-world deployment.

5. Multi-Agent Systems for Crisis Coordination

Multi-agent systems have been proposed as a solution for coordinating multiple stakeholders during disaster situations. Devadasu et al. developed a multi-agent learning model that supports adaptive crisis management in urban environments. In this model, multiple intelligent agents collaborate to collect data, analyze situations, and assist in decision-making processes. Multi-agent systems are particularly effective in dynamic disaster environments because they allow different agents to perform specialized tasks simultaneously. However, many of these systems focus mainly on response coordination and do not integrate early warning mechanisms or recovery monitoring.

6. IoT-Based Disaster Monitoring Systems

The Internet of Things (IoT) has introduced new opportunities for real-time disaster monitoring. Ray et al. proposed an IoT-based early warning system that uses sensor networks to monitor environmental conditions such as temperature, humidity, and water levels. The collected data is analysed in real time to detect potential disasters and generate early warning alerts. IoT-based systems improve situational awareness and enable authorities to respond quickly to disaster events. However, these systems generally focus only on monitoring and alert generation without addressing resource allocation or recovery tracking.

2.2 DEFINITION OF PROBLEM STATEMENT

Disaster management systems continue to face several critical challenges that reduce the effectiveness of emergency response and recovery operations. In many situations, disaster alerts are issued too late due to the lack of real-time monitoring and predictive capabilities, which leads to delayed evacuation and increased risks to human life and property. As disasters become more frequent and unpredictable, the need for faster and more reliable warning systems has become increasingly important.

Another major issue in disaster management is the inefficient allocation and distribution of emergency supplies. Limited visibility of available resources, manual planning methods, and poor coordination often result in shortages in some areas and wastage in others. Emergency responders may struggle to deliver essential supplies such as food, water, and medical aid on time, especially when accurate information about affected locations is not readily available.

Communication gaps among government agencies, emergency responders, volunteers, and affected communities further complicate disaster response efforts. Fragmented communication systems can cause confusion, delays in decision-making, and duplication of tasks during critical situations. In addition, many existing disaster management systems lack proper mechanisms to monitor post-disaster recovery activities, making it difficult for authorities to track rehabilitation progress and evaluate the effectiveness of relief operations.

Therefore, the main problem addressed in this project is the absence of a centralized and intelligent system that can integrate early warning, resource allocation, and recovery tracking into a single unified platform. The primary goal of this project is to develop a smart crisis management system that uses predictive analytics and real-time sensor data to provide timely alerts, optimize the distribution of emergency resources, and continuously monitor recovery efforts. By solving these problems, the system aims to improve response efficiency, reduce human and economic losses, enhance coordination among stakeholders, and strengthen community resilience during disaster situations.

2.3 EXISTING SYSTEM

Current disaster management systems rely heavily on a combination of historical data analysis, manual reporting methods, and basic software-based decision support tools to manage emergency situations. These traditional systems typically use past disaster records, demographic statistics, and geographic information to identify risk-prone areas and plan emergency preparedness activities. Authorities analyze previous incidents to estimate possible

disaster scenarios and determine response strategies. Although these methods provide useful background information, they often depend on outdated data and do not fully support real-time decision-making during rapidly changing disaster conditions.

Many existing platforms incorporate predictive models such as regression analysis, decision trees, and Random Forest algorithms to forecast disaster severity, identify vulnerable regions, and estimate resource requirements. Centralized dashboards are sometimes used to display information about disaster events, monitor affected areas, and track ongoing relief operations. However, these dashboards usually rely on periodic data updates rather than continuous real-time inputs, which can limit their effectiveness in urgent situations where immediate response is required.

In addition, several systems implement partial automation features, such as generating alerts when predefined thresholds—like rainfall levels or temperature changes—are exceeded. Some systems also use static optimization techniques to allocate emergency resources based on fixed rules or predefined plans. While these approaches improve efficiency to some extent, they lack flexibility and adaptability to unexpected or complex disaster scenarios. As a result, decision-making may still require significant manual intervention, which can slow down response times and increase the chances of human error.

Post-disaster recovery tracking in existing systems is often handled manually or through separate, disconnected tools, leading to fragmented data management and limited visibility of recovery progress. This separation makes it difficult for authorities to coordinate activities, monitor resource usage, and evaluate the overall effectiveness of relief efforts. Overall, although current disaster management systems provide essential support for emergency planning and response, they are mostly reactive rather than proactive and lack full integration of real-time analytics, automated coordination, and unified monitoring capabilities needed for efficient and modern crisis management.

Limitations of Existing Systems

Despite advancements, existing disaster management systems face several limitations

- **Delayed Response:** Manual reporting and fragmented coordination cause slow alerts and response times, reducing effectiveness during critical situations.
- **Inefficient Resource Allocation:** Resource distribution is often planned statically, without dynamic optimization based on evolving disaster conditions.
- **Limited Integration:** Early warning, emergency supply allocation, and recovery tracking are managed separately, leading to inefficiencies and poor coordination.

- **Scalability Issues:** Many platforms struggle to handle large-scale disasters or multiple simultaneous events due to software or data processing limitations.
- **Lack of Real-Time Monitoring:** Existing systems rely heavily on historical data and pre-defined rules, limiting their ability to respond dynamically to ongoing disasters.
- **High Dependence on Manual Processes:** Many disaster management activities, such as data entry, reporting, and decision-making, require manual intervention, increasing workload and the risk of human error.
- **Communication Delays:** Inefficient communication channels between authorities, responders, and communities can lead to misinformation, confusion, and delayed coordination during emergencies.
- **Poor Data Accuracy and Consistency:** Incomplete, outdated, or inconsistent data can result in incorrect assessments and ineffective response planning.
- **Difficulty in Tracking Recovery Progress:** Monitoring post-disaster rehabilitation and relief distribution is often done manually or through disconnected systems, making it hard to evaluate progress and performance.
- **Insufficient Transparency and Accountability:** Lack of centralized monitoring makes it difficult to track resource usage and operational activities, which can reduce trust among stakeholders.
- **Limited Accessibility for Remote Areas:** Existing systems may not effectively support communication or data collection in rural or remote regions with poor connectivity.
- **Higher Operational Costs:** Inefficient planning, duplicated efforts, and resource wastage can increase the overall cost of disaster management operations.

These challenges highlight the need for a unified, predictive analytics-driven platform that can provide end-to-end disaster management, including early warning, optimized emergency supply allocation, and post-disaster recovery tracking, improving both response speed and decision-making efficiency.

2.4 PROPOSED SYSTEM

The proposed system introduces a Smart Crisis Management Platform designed to provide a comprehensive and integrated solution for managing disasters from early detection to recovery. Unlike traditional systems that operate separately for different stages of disaster management, this platform combines early warning, emergency response, resource allocation, and recovery tracking into a single unified system. The system aims to improve preparedness,

reduce response time, and enhance coordination among authorities, emergency responders, volunteers, and affected communities through the use of modern digital technologies.

One of the key components of the proposed system is its real-time early warning capability, which uses predictive analytics and continuous data collection from IoT sensors, weather services, and geographic information systems (GIS). The system analyzes environmental and situational data to identify potential disaster zones and forecast the possible impact of events such as floods, earthquakes, fires, or storms. Based on this analysis, timely alerts and notifications are automatically generated and delivered to relevant stakeholders, enabling faster evacuation planning and preventive actions to minimize damage and protect lives.

Another important feature of the proposed system is the automated allocation of emergency resources, which ensures that essential supplies such as food, water, medical kits, and rescue equipment are distributed efficiently to affected areas. The platform evaluates factors such as disaster severity, population density, resource availability, and accessibility of locations to prioritize resource distribution. This intelligent allocation process reduces delays, prevents resource wastage, and ensures that aid reaches the most critical locations at the right time. In addition, the system continuously monitors resource usage and availability to support better planning and decision-making during emergencies.

Furthermore, the system includes a post-disaster recovery tracking mechanism that allows authorities to monitor rehabilitation activities, infrastructure restoration, and relief distribution in real time. A centralized dashboard provides a unified and visual representation of all crisis management activities, including incident reports, volunteer deployment, resource status, and recovery progress. The proposed platform is also designed with scalability in mind, enabling it to process large volumes of data from multiple sources simultaneously and support multiple disaster scenarios at the same time. This scalability and real-time processing capability significantly enhance overall responsiveness, operational efficiency, and the effectiveness of disaster management efforts.

Advantages of the Proposed System

The proposed system significantly improves upon the existing approaches by addressing key limitations:

- **Improved Accuracy:** Predictive models and real-time data reduce errors in disaster assessment.

- **Enhanced Coordination:** Unified platform enables better communication between multiple agencies.
- **Efficient Resource Allocation:** Optimizes supply distribution and reduces wastage.
- **Real-Time Monitoring:** Supports dynamic decision-making during crises.
- **Scalable and Robust:** Can handle large-scale disasters and multiple simultaneous events.
- **Faster Response Time:** Automated alerts and decision-support mechanisms enable authorities to respond quickly to emergencies, reducing delays in rescue and relief operations.
- **Reduced Human Error:** Automation of data analysis, alert generation, and resource allocation minimizes manual mistakes and improves operational reliability.
- **Better Preparedness and Planning:** Historical data storage and analytics help authorities analyse past disasters and prepare more effective response strategies for future emergencies.
- **Transparency and Accountability:** Real-time tracking of resources and activities ensures clear visibility of operations, improving trust and accountability among stakeholders.
- **Cost-Effective Operations:** Optimized resource utilization and reduced wastage lower the overall operational costs associated with disaster management.
- **Improved Public Safety:** Early warning systems and efficient coordination help protect lives and reduce injuries during disasters.
- **Data-Driven Decision Making:** Advanced analytics provide actionable insights that support informed and strategic decisions during all phases of disaster management.
- **Centralized Data Management:** All disaster-related information is stored in a single system, making it easier to access, update, and analyze data when needed.
- **Support for Post-Disaster Recovery:** Continuous monitoring of rehabilitation activities helps ensure faster recovery and better long-term planning for affected communities.

2.5 OBJECTIVES

- **Develop a Unified Crisis Management System:** Integrate early warning, emergency supply allocation, and post-disaster recovery tracking.
- **Enhance Response Efficiency:** Reduce delays in disaster alerts and resource deployment.

- **Improve Accuracy of Disaster Predictions** – Use predictive analytics and IoT data for timely detection.
- **Ensure Scalability and Real-Time Processing** – Handle large-scale disasters efficiently.
- **Strengthen Community Resilience** – Minimize human and economic losses through proactive disaster management.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specify the logical characteristics of each interface between the software product and the hardware components of the system. The following are the hardware requirements:

- Processor : Intel Core i3 or higher
- RAM : 4 GB minimum
- Storage : 128 GB or higher SSD

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specify the logical characteristics of each interface and software components of the system. The following are the software requirements:

- Frontend : React Native / React.js
- Backend : Node.js with Express
- Database : Firebase Firestore

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

System architecture refers to the structural framework that defines how components of SCM interact, communicate, and collaborate to achieve the project's goals. A well-defined architecture ensures modularity, replaceability of components, and traceability of decisions, making the system maintainable and extensible for future enhancements.

3.1 PROJECT ARCHITECTURE

The architecture of the Smart Crisis Management Platform outlines the end-to-end workflow, how different components interact to manage emergencies. Users such as Citizens, Rescue Teams, and Administrators access the system through mobile or web applications. All requests pass through an API Gateway, which handles authentication and secure communication. The Business Logic Layer includes modules like Disaster Detection, Alert System, Request Handler, Resource Manager, and AI Predictor to process incidents and manage resources. External systems such as Sensors/IoT, Social-Media, and Government APIs provide real-time data to detect emergencies. The system stores all information in the Main Database and File Storage. Overall, the architecture ensures efficient communication, quick response, and reliable crisis management.

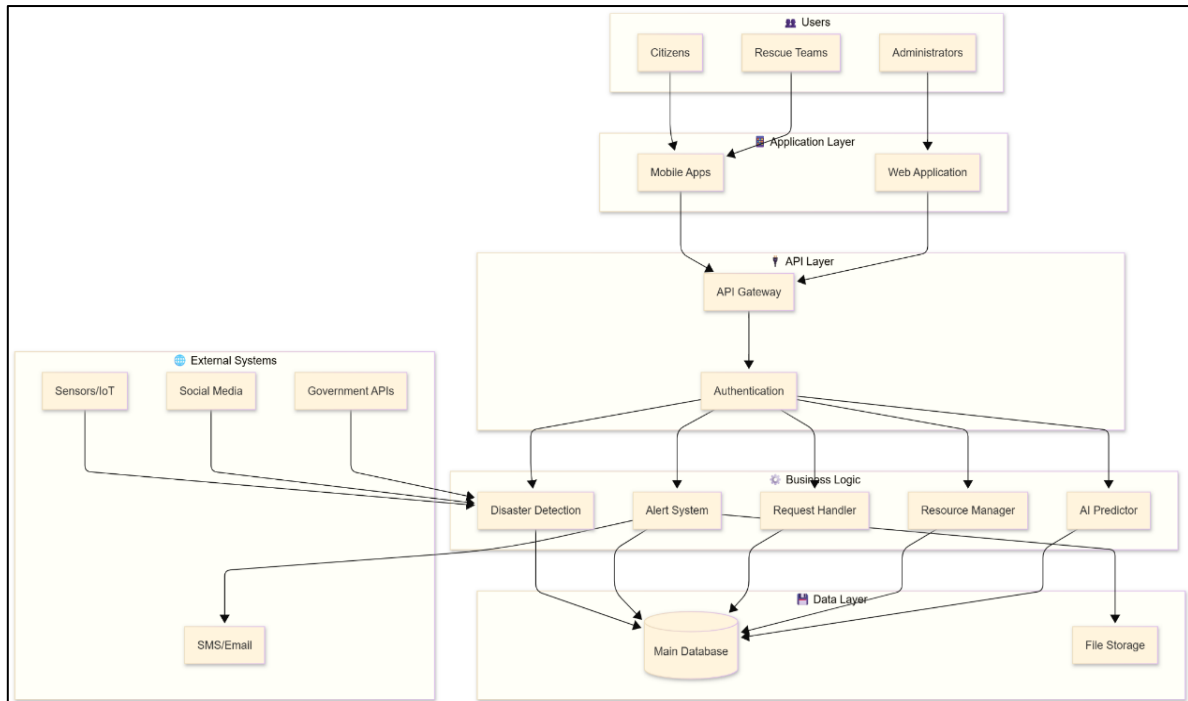


Figure 3.1: Project Architecture of Smart Crisis Management: A Unified System for Early Warning, Emergency Supply Allocation, and Post-Disaster Recovery Tracking

3.2 DESCRIPTION

Input Data: The system uses historical disaster datasets, including natural and human-made disaster records, demographic data, and geographic information relevant to affected areas.

Data Preprocessing: Collected data is cleaned, normalized, and structured to ensure compatibility with predictive analytics algorithms. Missing or inconsistent entries are handled to improve model reliability.

Predictive Analytics: Machine learning and statistical models analyze historical and current disaster data to forecast potential risk zones, severity, and resource requirements. This step allows the system to generate early warnings and prioritize affected areas.

Resource Allocation Module: Based on predictive insights, the system automatically determines the optimal allocation of emergency supplies, considering factors such as severity, population density, and accessibility of affected regions.

Post-Disaster Recovery Tracking: The system monitors recovery efforts, tracking resource utilization, rehabilitation progress, and response efficiency. Reports are generated to support decision-making and improve future disaster management strategies.

Centralized Dashboard: A unified dashboard visualizes all disaster management activities, including alerts, resource distribution, and recovery status. Authorities and responders can access real-time updates and coordinate actions effectively.

Training and Evaluation: Predictive models are validated using historical disaster scenarios, measuring accuracy, response time improvement, and resource optimization. This ensures the system provides reliable, actionable outputs for decision-makers.

3.3 UML DIAGRAMS

Unified Modelling Language (UML) is a standardized visual language used to model, design, and document the architecture of software systems. It provides a set of graphical notations to represent the structure and behaviour of a system, making complex systems easier to understand and communicate among developers, stakeholders, and business analysts.

3.3.1 CLASS DIAGRAM

A Class Diagram represents the static structure of the Smart Crisis Management System by illustrating its main classes, their attributes, methods, and relationships. It includes key components such as Disaster, User, Alert, HelpRequest, Shelter, RescueTeam, and Resource, each responsible for specific functionalities like alert generation, request handling, user management, and resource allocation. The diagram shows how a disaster triggers alerts and causes help requests, users create requests and access shelters, and rescue teams are assigned to handle requests using available resources. This structured view helps developers understand how different parts of the system interact, ensuring efficient coordination, clear responsibility distribution, and better system design for managing emergency situations.

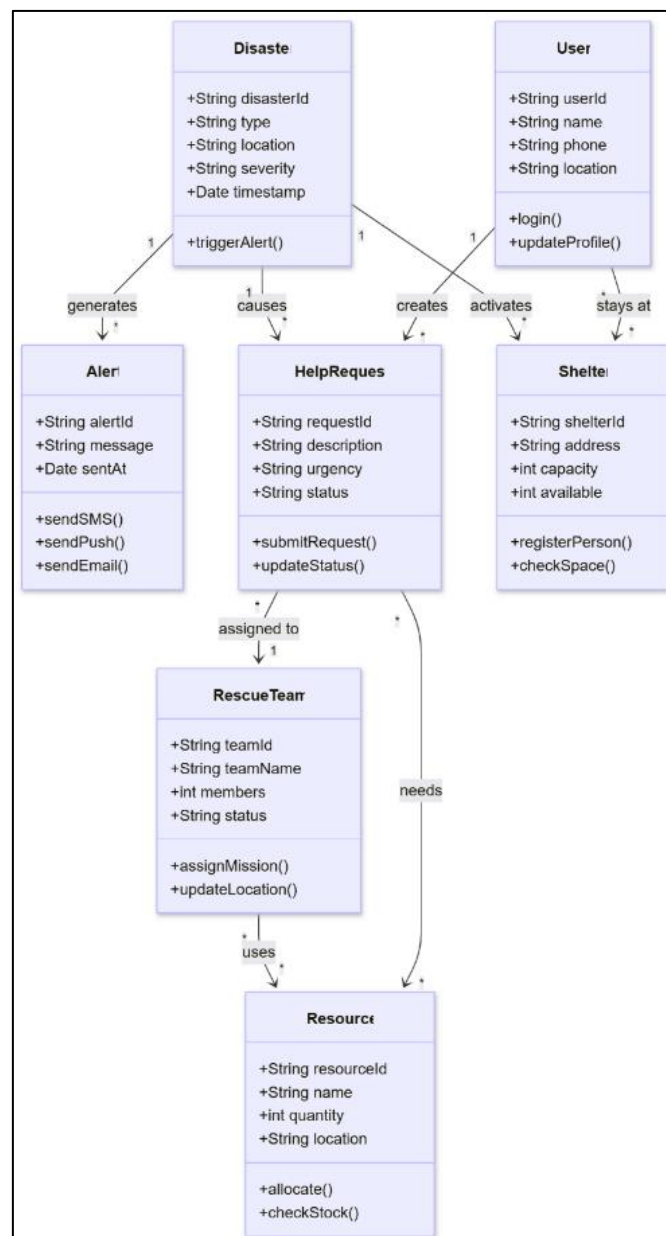


Figure 3.2: Class Diagram of Smart Crisis Management System

3.3.2 USE CASE DIAGRAM

The Use Case Diagram for the Smart Crisis Management System represents how three main actors—Citizen, Volunteer, and Admin—interact with the system to manage emergency situations efficiently. The dotted rectangle shows the system boundary, which contains all the functions provided by the system. The Citizen can register or log in, report incidents, view incident status, and receive alerts about emergencies. The Volunteer can log in to the system, view incident status, receive alerts, and check assigned tasks to assist in handling emergencies. The Admin plays a major role in controlling and coordinating the system by sending alerts, assigning volunteers, allocating resources, updating incident status, generating early warnings, and managing users. Overall, the diagram illustrates the flow of communication and responsibilities among users, ensuring that incidents are reported, managed, and resolved in an organized and timely manner within the Smart Crisis Management System.

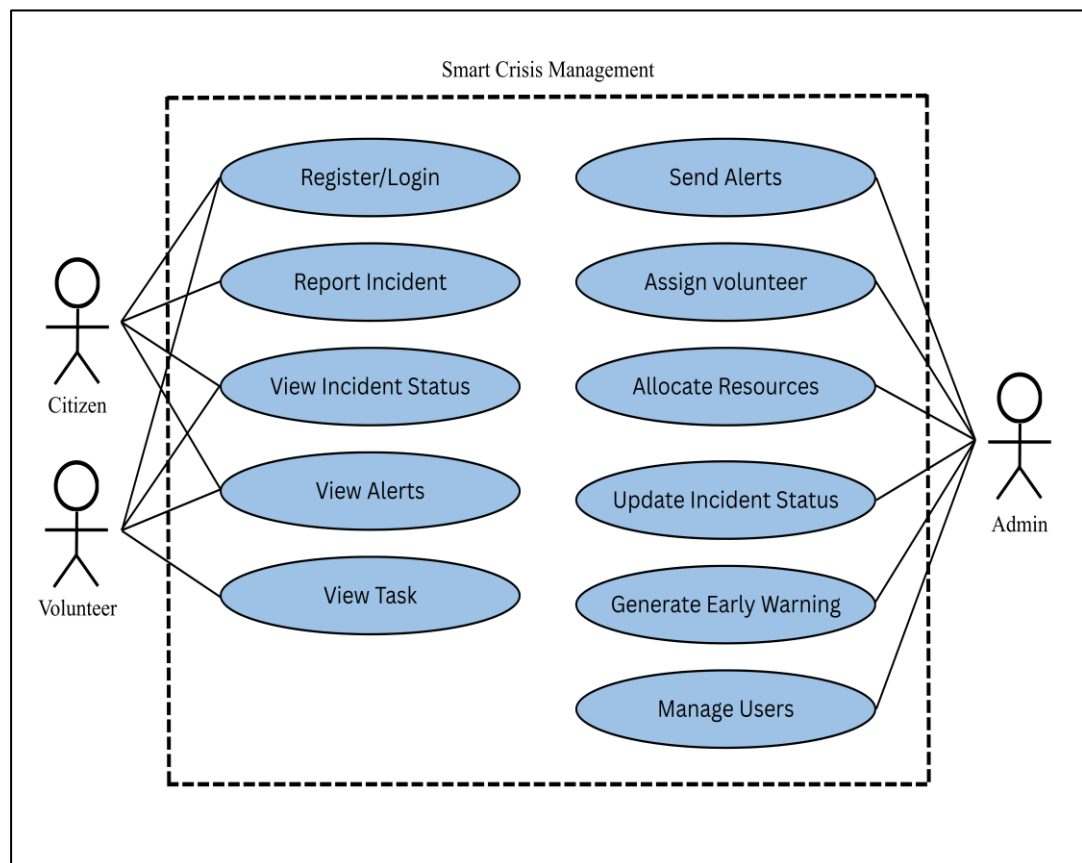


Figure 3.3: Use Case Diagram of Smart Crisis Management System

3.3.3 SEQUENCE DIAGRAM

The Sequence Diagram illustrates the dynamic flow of interactions within the Smart Crisis Management System over time, showing how different components communicate during a disaster scenario. It includes key entities such as Crisis Manager, Early Warning System, Emergency Supply System, Post-Disaster Recovery Tracker, Data Analytics Service, Communication System, Emergency Response Team, Local Authorities, and the Public. The process begins with monitoring environmental data and analyzing potential threats, followed by generating and sending early warning alerts through the communication system. Authorities and response teams receive alerts, notify the public, and initiate emergency actions such as deploying teams and allocating supplies. The system then tracks recovery progress, updates the public on the situation, and collects feedback for further improvement. This diagram helps in understanding the sequence of operations, message flow, and coordination among system components to ensure an efficient and timely disaster response.

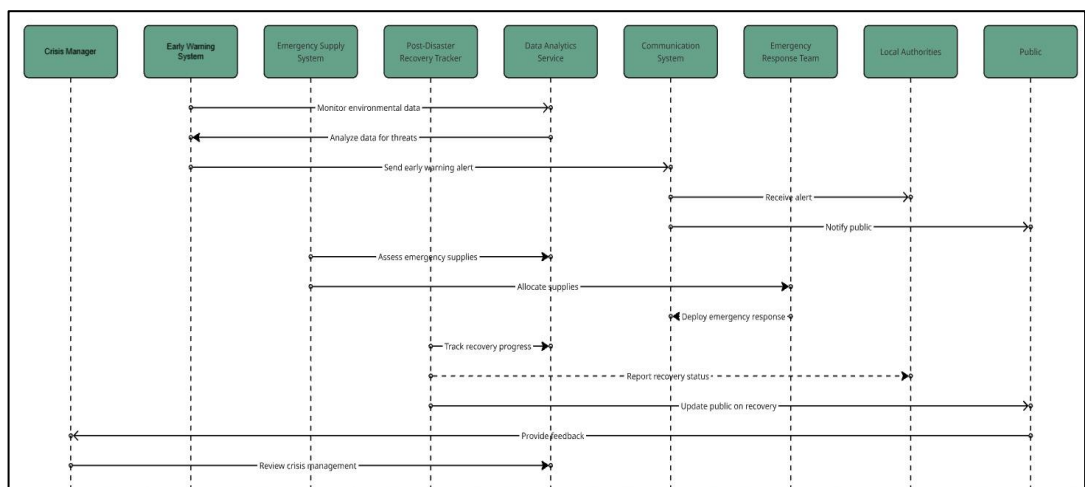


Figure 3.4: Sequence Diagram of Smart Crisis Management System

3.3.4 ACTIVITY DIAGRAM

The Activity Diagram represents the workflow of resource management in the Smart Crisis Management System, showing the sequence of actions and decision-making steps involved during emergency operations. The process begins with monitoring and assessing the situation to evaluate resource availability. If a shortage is identified, the system proceeds to request additional resources and procure them; otherwise, it directly moves to allocation. Once resources are allocated, they are dispatched and delivered to the required location. Finally, the process ends with confirmation of successful delivery. This diagram helps in understanding the step-by-step flow of activities, decision points, and control logic, ensuring efficient resource distribution and timely response during crisis situations.

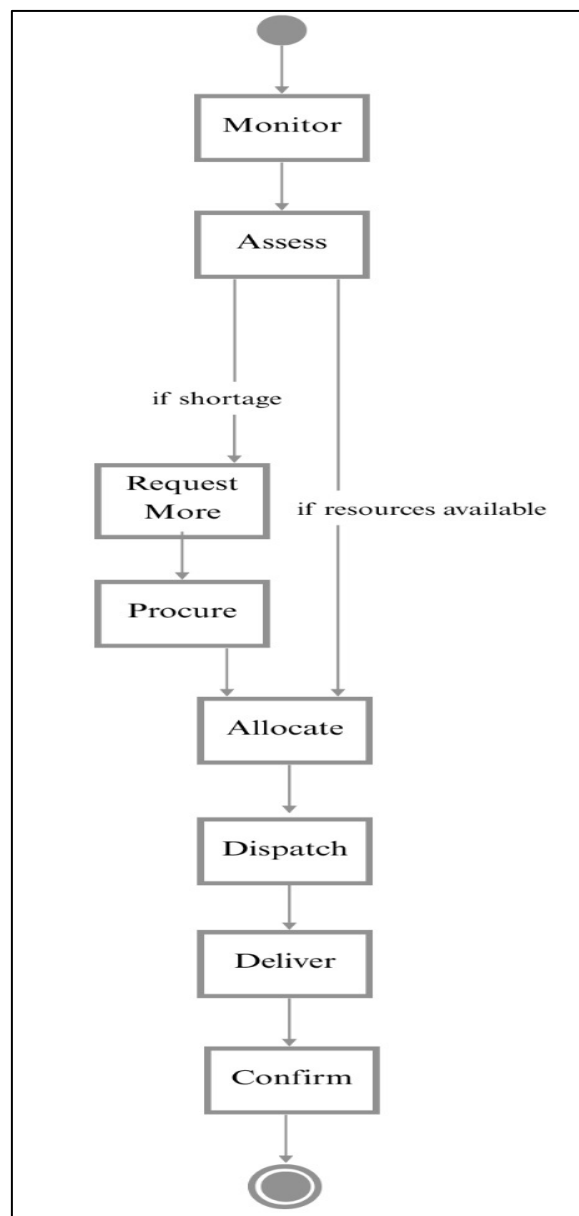


Figure 3.5: Activity Diagram of Smart Crisis Management System

3.3.5 COMPONENT DIAGRAM

The Component Diagram represents the high-level architecture of the Smart Crisis Management System by showing how different system components are organized and interact with each other. It includes the User Interface layer with Mobile App and Web Portal, which communicate with the core system through an API Gateway. The core system consists of key services such as Alert Service, Request Service, Resource Service, and Map Service, each handling specific functionalities like notifications, help requests, resource management, and location tracking. These services interact with the data storage layer, which includes a database and cache for efficient data handling. Additionally, the system integrates with external services such as SMS Gateway, Weather API, and Social Media to enhance communication and real-time data processing. This diagram helps developers understand the modular structure, data flow, and integration points, ensuring scalability, maintainability, and efficient system performance.

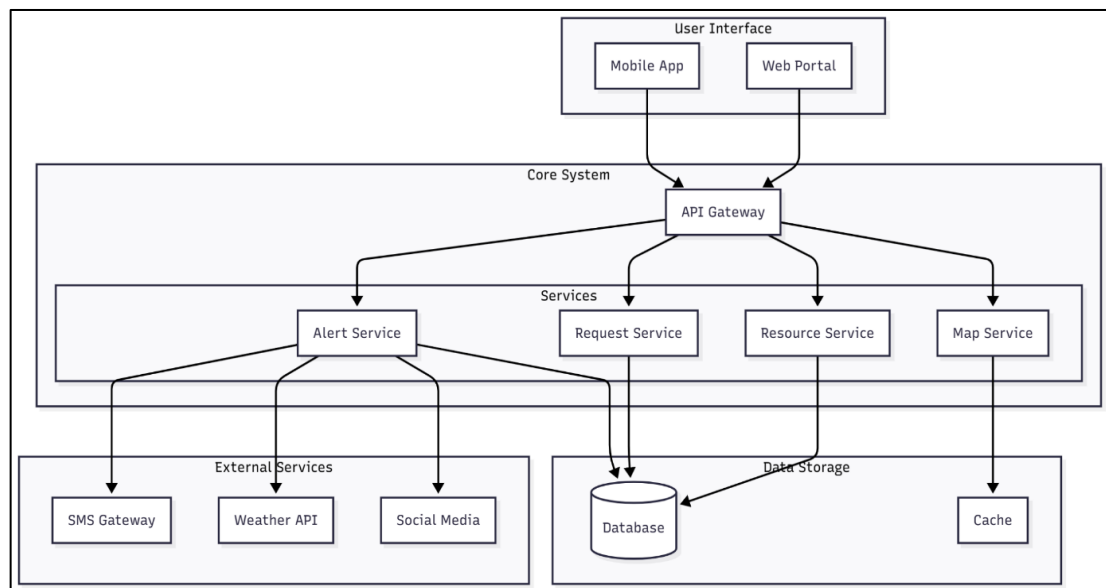


Figure 3.6: Component Diagram of Smart Crisis Management System

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of a project involves executing the planned strategies and tasks. It requires meticulous coordination, resource allocation, and monitoring to ensure that objectives are met efficiently. Effective implementation is crucial for achieving project goals and delivering expected outcomes within the set timeline and budget constraints.

4.1 ALGORITHMS USED

4.1.1. Machine Learning-Based Risk Prediction Algorithm:

Machine learning algorithms are used in the system to analyze disaster-related data and predict potential crisis situations. These algorithms process historical disaster data, environmental conditions, and real-time sensor data to identify patterns that indicate possible disasters.

By using supervised learning techniques, the system can classify disaster risks such as floods, earthquakes, fires, or storms. This allows the system to generate early warning alerts before the disaster causes severe damage.

Advantages:

- Helps identify disaster risks at an early stage.
- Improves decision-making by analysing historical and real-time data.
- Supports automated prediction without human intervention.

Disadvantages:

- Requires large datasets for accurate predictions.
- Prediction accuracy depends on the quality of training data.

4.1.2. Multi-Agent System (MAS):

Multi-Agent System consists of multiple intelligent agents that work together to perform different tasks within the disaster management system. Each agent is responsible for a specific function such as data collection, risk analysis, resource allocation, or communication with authorities.

In the Smart Crisis Management System, agents collaborate to analyze disaster conditions and coordinate response activities. For example, one agent collects sensor data, another agent analyses disaster severity, and another agent manages emergency supply allocation.

Advantages:

- Allows distributed decision-making.
- Improves system scalability and flexibility.
- Enables coordination between different system components.

Disadvantages:

- Designing agent communication protocols can be complex.
- Requires proper synchronization between agents.

4.1.3. Resource Allocation Optimization Algorithm:

During disaster response operations, efficient allocation of emergency resources such as rescue teams, medical supplies, and transportation is essential. Optimization algorithms are used in the system to determine the best way to distribute available resources based on disaster severity, location, and urgency.

The algorithm evaluates multiple factors such as distance, available resources, and affected population to allocate supplies effectively.

Advantages:

- Ensures efficient distribution of emergency resources.
- Reduces response time during disasters.
- Helps prioritize high-risk areas

Disadvantages:

- May require complex calculations for large-scale disasters.
- Performance depends on real-time availability of resource data.

4.1.4. IoT-Based Data Collection Algorithm:

The system uses Internet of Things (IoT) technologies to collect real-time environmental data from sensors. These sensors monitor parameters such as temperature, humidity, water levels, and seismic activity.

The IoT data collection algorithm gathers sensor data and sends it to the central system for analysis. This data is used to detect abnormal environmental conditions and generate early warning alerts.

Advantages:

- Provides real-time monitoring of environmental conditions.

- Improves situational awareness during disasters.
- Enables automatic data collection from multiple sources

Disadvantages:

- Requires reliable network connectivity.
- Large numbers of sensors may increase system complexity.

4.2 SAMPLE CODE

4.2.1 Authentication Module - Login & Role Management

```
export enum UserRole {
  admin = "admin",
  volunteer = "volunteer",
  citizen = "citizen",
}
const SESSION_ROLE_KEY = "sc_session_role";
// Login Function (API / Database Based)
export const loginUser = async (
  username: string,
  password: string
): Promise<UserRole | null> => {
  try {
    const response = await fetch("http://localhost:5000/api/login", {
      method: "POST",
      headers: {
        "Content-Type": "application/json",
      },
      body: JSON.stringify({ username, password }),
    });
    if (!response.ok) {
      throw new Error("Invalid credentials");
    }
    const data = await response.json();
    // Expected response: { role: "admin" | "volunteer" | "citizen" }
    const role: UserRole = data.role;
    // Save session
    sessionStorage.setItem(SESSION_ROLE_KEY, role);
```

```
// Save login log
saveLoginLog(username, role);
return role;
} catch (error) {
  console.error("Login failed:", error);
  return null;
}
};

// Logout Function
export const logoutUser = () => {
  sessionStorage.removeItem(SESSION_ROLE_KEY);
};

// Get Current Logged-in Role
export const getCurrentUserRole = (): UserRole | null => {
  const role = sessionStorage.getItem(SESSION_ROLE_KEY);
  if (role === UserRole.admin) return UserRole.admin;
  if (role === UserRole.volunteer) return UserRole.volunteer;
  if (role === UserRole.citizen) return UserRole.citizen;
  return null;
};

// Save Login Log
export const saveLoginLog = (username: string, role: UserRole) => {
  const logs = JSON.parse(localStorage.getItem("loginLogs") || "[]");
  logs.push({
    username,
    role,
    loginTime: new Date().toISOString(),
  });
  localStorage.setItem("loginLogs", JSON.stringify(logs));
};
```

4.2.2 Alert Management Module

```
export interface Alert {
  id: number;
  message: string;
  status: string;
}
```

```
let alerts: Alert[] = [];  
export const createAlert = (message: string) => {  
  const newAlert: Alert = {  
    id: alerts.length + 1,  
    message,  
    status: "Active",  
  };  
  alerts.push(newAlert);  
};  
export const getAlerts = () => {  
  return alerts;  
};
```

4.2.3 Volunteer Management Module

```
import React, { useEffect, useState } from "react";  
interface Volunteer {  
  id: number;  
  name: string;  
  email: string;  
  phone: string;  
  status: string;  
}  
const VolunteerManagementModule = () => {  
  const [volunteers, setVolunteers] = useState<Volunteer[]>([]);  
  const [newVolunteer, setNewVolunteer] = useState({  
    name: "",  
    email: "",  
    phone: "",  
  });  
  // Fetch volunteers from database  
  useEffect(() => {  
    fetchVolunteers();  
  }, []);  
  const fetchVolunteers = async () => {  
    try {  
      const response = await fetch(  
        "http://localhost:5000/api/volunteers"
```

```
);
const data = await response.json();
setVolunteers(data);
} catch (error) {
  console.error("Error fetching volunteers:", error);
}
};

// Handle input change
const handleChange = (
  e: React.ChangeEvent<HTMLInputElement>
) => {
  setNewVolunteer({
    newVolunteer,
    [e.target.name]: e.target.value,
  });
};

// Add volunteer to database
const addVolunteer = async () => {
  try {
    await fetch(
      "http://localhost:5000/api/volunteers",
      {
        method: "POST",
        headers: {
          "Content-Type":
            "application/json",
        },
        body: JSON.stringify(
          newVolunteer
        ),
      }
    );
    fetchVolunteers();
    setNewVolunteer({
      name: "",
      email: "",
    });
  }
};
```

```
    phone: "",
  });
} catch (error) {
  console.error(
    "Error adding volunteer:",
    error
  );
}
};

// Remove volunteer
const removeVolunteer = async (
  id: number
) => {
  try {
    await fetch(
      `http://localhost:5000/api/volunteers/${id}`,
      {
        method: "DELETE",
      }
    );
    fetchVolunteers();
  } catch (error) {
    console.error(
      "Error removing volunteer:",
      error
    );
  }
};

return (
  <div>
    <h2>Volunteer Management</h2>
    { /* Add Volunteer Form */ }
    <div>
      <h3>Add Volunteer</h3>
      <input
        type="text"
```

```
name="name"
placeholder="Name"
value={new Volunteer.name}
onChange={handleChange}
/>
<input
  type="email"
  name="email"
  placeholder="Email"
  value={new Volunteer.email}
  onChange={handleChange}
/>
<input
  type="text"
  name="phone"
  placeholder="Phone"
  value={new Volunteer.phone}
  onChange={handleChange}
/>
<button onClick={addVolunteer}>
  Add Volunteer
</button>
</div>
{/* Volunteer List */}
<table border={1} cellPadding={8}>
  <thead>
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Email</th>
      <th>Phone</th>
      <th>Status</th>
      <th>Action</th>
    </tr>
  </thead>
  <tbody>
```

```

{volunteers.map(
  (volunteer) => (
    <tr key={volunteer.id}>
      <td>{volunteer.id}</td>
      <td>{volunteer.name}</td>
      <td>{volunteer.email}</td>
      <td>{volunteer.phone}</td>
      <td>{volunteer.status}</td>
      <td>
        <button
          onClick={() =>
            removeVolunteer(
              volunteer.id
            )
          }
        >
          Remove
        </button>
      </td>
    </tr>
  )
)}
</tbody>
</table>
</div>
);
};
export default VolunteerManagementModule;

```

4.2.4 Resource Availability Module

```

export interface Resource {
  id: number;
  name: string;
  quantity: number;
  status: string;
}
let resources: Resource[] = [];

```

```
export const addResource = (name: string, quantity: number) => {
  const status = quantity > 10 ? "Available" : "Low";
  resources.push({
    id: resources.length + 1,
    name,
    quantity,
    status,
  });
};

export const getResources = () => {
  return resources;
};
```

4.2.5 Main Application Module

```
import DashboardLayout from "../components/DashboardLayout";
import ThemeToggle from "../components/ThemeToggle";
function App() {
  return (
    <div>
      <DashboardLayout>
        <h1>Smart Crisis Management System</h1>
        <ThemeToggle />
      </DashboardLayout>
    </div>
  );
}

export default App;
```

4.2.6 Admin Dashboard Layout Module

```
import React from "react";
interface Props {
  children: React.ReactNode;
}

const AdminDashboardLayout = ({ children }: Props) => {
  return (
    <div className="dashboard-container">
      { /* Sidebar */ }
      <aside className="sidebar">
```

```
<h2>Admin Panel</h2>
<ul>
  <li>Dashboard</li>
  <li>Alerts</li>
  <li>Volunteers</li>
  <li>Login Logs</li>
  <li>Resource Availability</li>
  <li>Settings</li>
  <li>Logout</li>
</ul>
</aside>
{/* Main Content */}
<main className="main-content">
  {children}
</main>

</div>
);
};
export default AdminDashboardLayout;
4.2.7 Citizen Dashboard Layout Module
import React from "react";
interface Props {
  children: React.ReactNode;
}
const CitizenDashboardLayout = ({ children }: Props) => {
  return (
    <div className="dashboard-container">
      {/* Sidebar */}
      <aside className="sidebar">
        <h2>Citizen Panel</h2>
        <ul>
          <li>Dashboard</li>
          <li>Report Incident</li>
          <li>View Alerts</li>
          <li>My Reports</li>
        </ul>
      </aside>
      {children}
    </div>
  );
};
export default CitizenDashboardLayout;
```

```
      <li>Settings</li>
      <li>Logout</li>
    </ul>
  </aside>
  { /* Main Content */ }
  <main className="main-content">
    {children}
  </main>
</div>
);
};
export default CitizenDashboardLayout;

```

4.2.8 Volunteer Dashboard Layout Module

```
import React from "react";
interface Props {
  children: React.ReactNode;
}
const VolunteerDashboardLayout = ({ children }: Props) => {
  return (
    <div className="dashboard-container">
      { /* Sidebar */ }
      <aside className="sidebar">
        <h2>Volunteer Panel</h2>
        <ul>
          <li>Dashboard</li>
          <li>My Tasks</li>
          <li>Assigned Incidents</li>
          <li>Update Status</li>
          <li>Settings</li>
          <li>Logout</li>
        </ul>
      </aside>
      { /* Main Content */ }
      <main className="main-content">
        {children}
      </main>
    </div>
  );
};

```

</div>

);

};

export default VolunteerDashboardLayout;

4.2.9 My Tasks Module (Volunteer)

import React, { useState } from "react";

interface Task {

id: number;

taskTitle: string;

location: string;

status: string;

}

const MyTasksModule = () => {

// Sample task list (fetched from database in real system)

const [tasks, setTasks] = useState<Task[]>([

{

id: 1,

taskTitle: "Rescue Operation",

location: "City Center",

status: "Pending",

},

{

id: 2,

taskTitle: "Medical Assistance",

location: "Community Hall",

status: "In Progress",

},

]);

// Update Task Status

const updateStatus = (taskId: number) => {

const updatedTasks = tasks.map((task) =>

task.id === taskId

? { ...task, status: "Completed" }

: task

);

setTasks(updatedTasks);

```
};  
return (  
  <div>  
    <h2>My Tasks</h2>  
  
    <table border={1} cellPadding={8}>  
      <thead>  
        <tr>  
          <th>Task ID</th>  
          <th>Task Title</th>  
          <th>Location</th>  
          <th>Status</th>  
          <th>Action</th>  
        </tr>  
      </thead>  
      <tbody>  
        {tasks.map((task) => (  
          <tr key={task.id}>  
            <td>{task.id}</td>  
            <td>{task.taskTitle}</td>  
            <td>{task.location}</td>  
            <td>{task.status}</td>  
            <td>  
              <button  
                onClick={() =>  
                  updateStatus(task.id)  
                }  
              >  
                Mark Completed  
              </button>  
            </td>  
          </tr>  
        ))}  
      </tbody>  
    </table>  
  </div>
```

```
);  
};  
export default MyTasksModule;  
4.2.10 Report Incident Module (Citizen)  
import React, { useState } from "react";  
interface Incident {  
  title: string;  
  description: string;  
  location: string;  
  date: string;  
}  
const ReportIncidentModule = () => {  
  const [incident, setIncident] = useState<Incident>({  
    title: "",  
    description: "",  
    location: "",  
    date: "",  
  });  
  // Handle input change  
  const handleChange = (  
    e: React.ChangeEvent<HTMLInputElement | HTMLTextAreaElement>  
  ) => {  
    setIncident({  
      incident,  
      [e.target.name]: e.target.value,  
    });  
  };  
  // Submit incident  
  const handleSubmit = (  
    e: React.FormEvent  
  ) => {  
    e.preventDefault();  
    console.log("Incident Reported:", incident);  
    alert("Incident reported successfully!");  
    // Reset form  
    setIncident({
```

```
title: "",
description: "",
location: "",
date: "",
});
};
return (
  <div>
    <h2>Report Incident</h2>
    <form onSubmit={handleSubmit}>
      <div>
        <label>Incident Title:</label>
        <input
          type="text"
          name="title"
          value={incident.title}
          onChange={handleChange}
          required
        />
      </div>
      <div>
        <label>Description:</label>
        <textarea
          name="description"
          value={incident.description}
          onChange={handleChange}
          required
        />
      </div>
      <div>
        <label>Location:</label>
        <input
          type="text"
          name="location"
          value={incident.location}
          onChange={handleChange}
        />
      </div>
    </form>
  </div>
);
```

```
        required
      />
    </div>
    <div>
      <label>Date:</label>
      <input
        type="date"
        name="date"
        value={incident.date}
        onChange={handleChange}
        required
      />
    </div>
    <button type="submit">
      Submit Report
    </button>
  </form>
</div>
);
};
export default ReportIncidentModule;
4.2.11 Main Application Module
import DashboardLayout from "../components/DashboardLayout";
import ThemeToggle from "../components/ThemeToggle";
function App() {
  return (
    <div>
      <DashboardLayout>
        <h1>Smart Crisis Management System</h1>
        <ThemeToggle />
      </DashboardLayout>
    </div>
  );
}
export default App;
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

5.1 SYSTEM PERFORMANCE AND EVALUATION SUMMARY

In the below screen, the Smart Crisis Management System home page is displayed. The interface allows users to log in according to their role in the system. The three main roles provided are Admin, Volunteer, and Citizen. Each role has a separate login option which directs the user to the respective dashboard.

The Admin can manage the entire system including incident monitoring, alert generation, resource allocation, and recovery tracking. The Volunteer can view assigned tasks and respond to emergency alerts. The Citizen can report incidents, request help, and receive notifications about disasters. This interface acts as the entry point of the system and enables role-based access to different functionalities of the Smart Crisis Management platform.

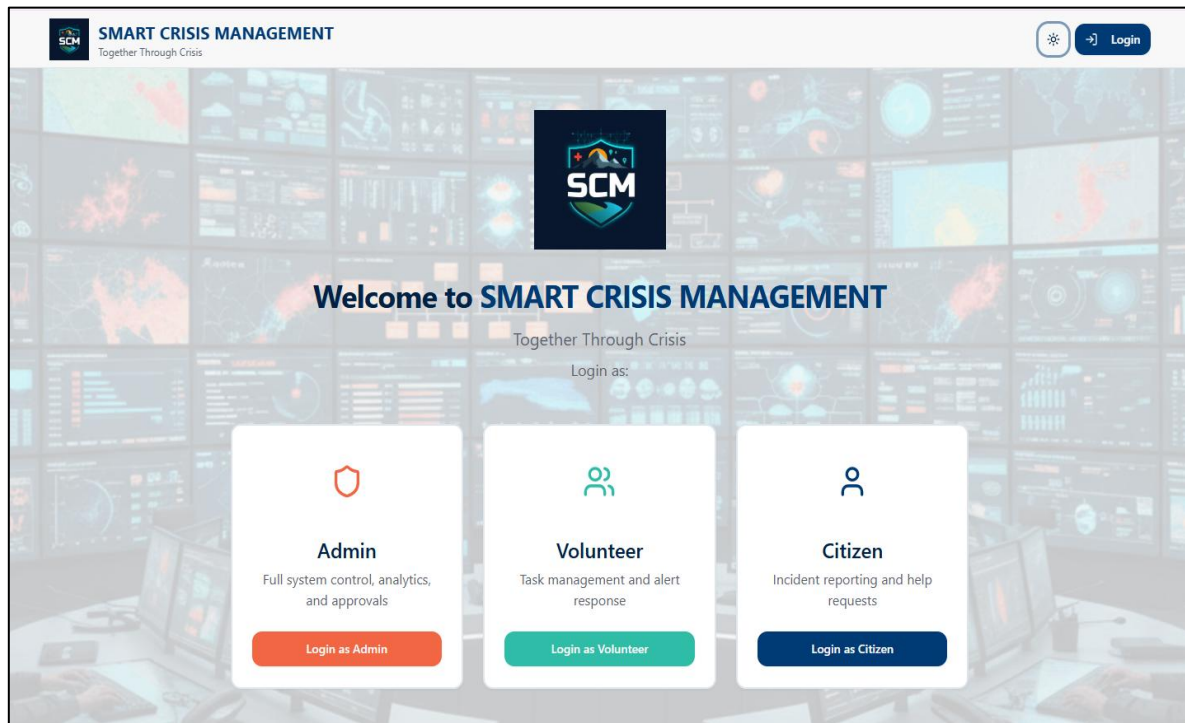


Figure 5.1: GUI/Main Interface of Smart Crisis Management System

5.2 ADMIN DASHBOARD:

In below screen, the Admin Dashboard of the Smart Crisis Management System is displayed. This dashboard provides the administrator with an overview of the entire crisis management platform. The admin can monitor important system information such as critical alerts, open incidents, active tasks, and active volunteers. These sections help administrators quickly understand the current status of emergencies and system activities.

The dashboard also displays resource status, including available food, water, medicine, vehicles, and personnel required for disaster response. In addition, the system information panel provides a summary of total incidents, alerts, tasks, volunteers, and the administrator role. The navigation menu allows the admin to access different modules such as Users, Volunteers, Incidents, Alerts, Tasks, Resources, Recovery, and Analytics. This interface helps administrators manage disaster response operations efficiently and coordinate resources during crisis situations.

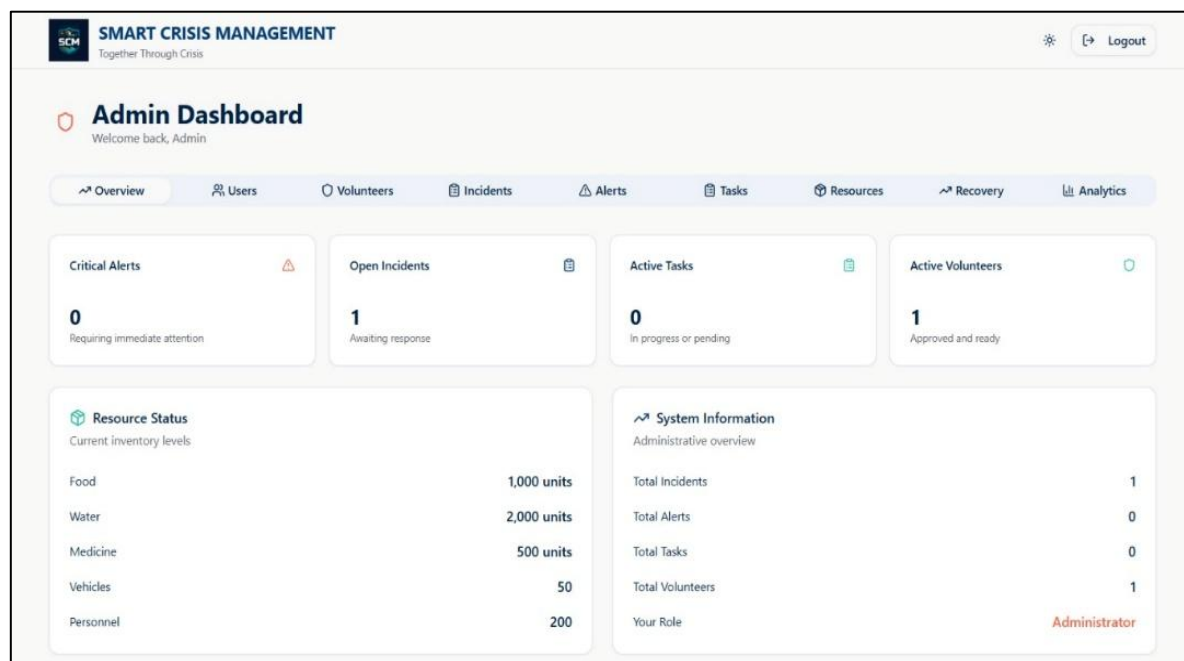


Figure 5.2: Admin Dashboard of Smart Crisis Management System

5.3 VOLUNTEER DASHBOARD

In the below screen, the Volunteer Dashboard of the Smart Crisis Management System is displayed. This interface allows volunteers to view and manage the tasks assigned to them during emergency situations. The dashboard provides information about assigned incidents, active tasks, completed tasks, and nearby incidents within a specific radius. These features help volunteers quickly understand their responsibilities and respond to emergencies efficiently.

The dashboard also includes navigation options such as Assigned Incidents, Nearby Incidents, Tasks, Recovery Updates, and Profile. Volunteers can monitor incidents assigned to them, track task progress, and view updates related to recovery activities. This interface helps volunteers coordinate with administrators and contribute effectively to disaster response and recovery operations.

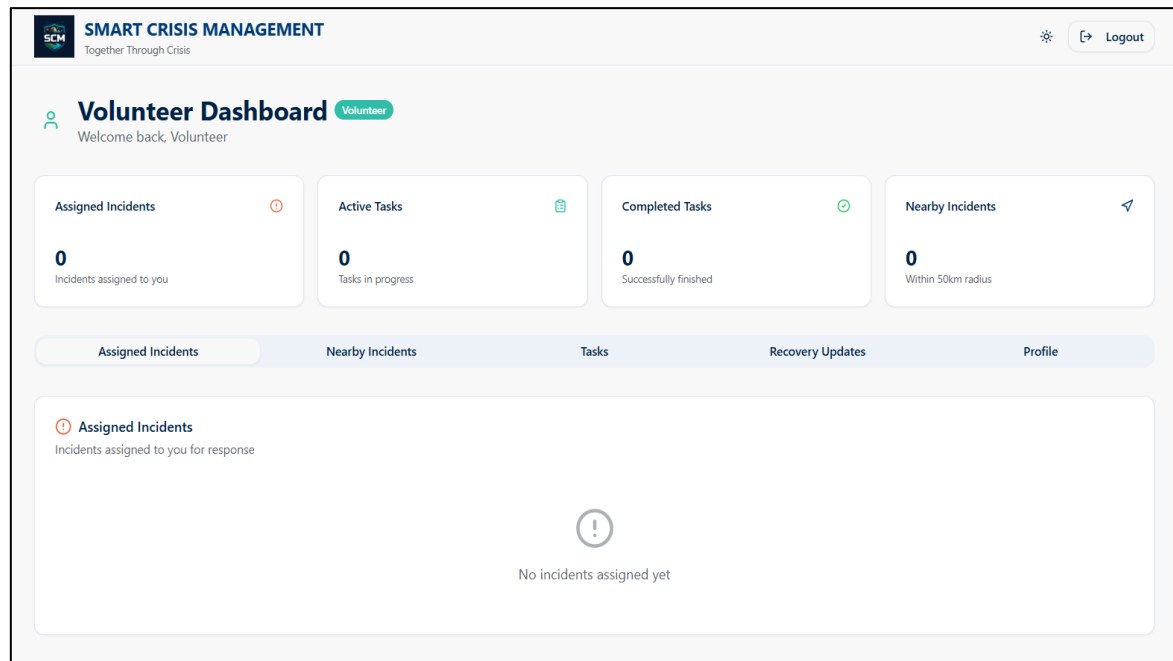


Figure 5.3: Volunteer Dashboard of Smart Crisis Management System

5.4 CITIZEN DASHBOARD

In below screen, the Citizen Dashboard of the Smart Crisis Management System is displayed. This interface allows citizens to interact with the system by reporting incidents and receiving important alerts during emergency situations. The dashboard provides an overview of early alerts, submitted incidents, rescue updates, and assistance requests, helping citizens stay informed about disaster-related activities.

The interface also includes a Report Incident section, where citizens can submit details about incidents they observe. Users can select the incident type, enter the location, provide a description of the situation, and upload images related to the incident. These reports are then sent to the system so that administrators and volunteers can take appropriate action. The dashboard also provides navigation options such as Nearby Alerts, Rescue Updates, and Profile, enabling citizens to track emergency updates and manage their information efficiently.

The screenshot displays the Citizen Dashboard of the Smart Crisis Management System. At the top, the header includes the SCM logo, the text "SMART CRISIS MANAGEMENT Together Through Crisis", a settings icon, and a "Logout" button. The main heading is "Citizen Dashboard" with a welcome message "Welcome back, Citizen". Below this, there are four summary cards: "Early Alerts" (0 Received), "Submitted Incidents" (0 Reports sent), "Rescue Updates" (0 Updates received), and "Assistance Requests" (0 Help requests sent). A navigation bar contains four tabs: "Report Incident", "Nearby Alerts", "Rescue Updates", and "Profile". The "Report Incident" tab is selected, showing a form to "Submit details about an incident you witnessed". The form has three main sections: "Incident Type" with a dropdown menu, "Location (Area / City)" with a text input field, and "Description" with a larger text input field. To the right of the form is a "System Information" section with a photo upload area that says "Drag & drop photos here or click to upload (Max 5 photos)". At the bottom right of the form are "Cancel" and "Submit Incident" buttons. The footer of the dashboard reads "Emergency Response • Resource Management • Recovery Tracking".

Figure 5.4: Citizen Dashboard of Smart Crisis Management System

5.5 SYSTEM PERFORMANCE METRICS

In below screen, the performance evaluation graph of the Smart Crisis Management System is shown. The bar chart represents the system performance based on different evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. These metrics are commonly used to measure the effectiveness and reliability of the system in handling disaster-related data and incident management.

From the graph, the system achieves high performance across all metrics, with Accuracy around 94%, Precision around 93.6%, Recall around 92.8%, and F1-Score around 93.2%. These results indicate that the Smart Crisis Management System performs efficiently in detecting incidents, generating alerts, and managing disaster response tasks. The evaluation demonstrates that the system provides reliable and accurate outputs for crisis management operations.

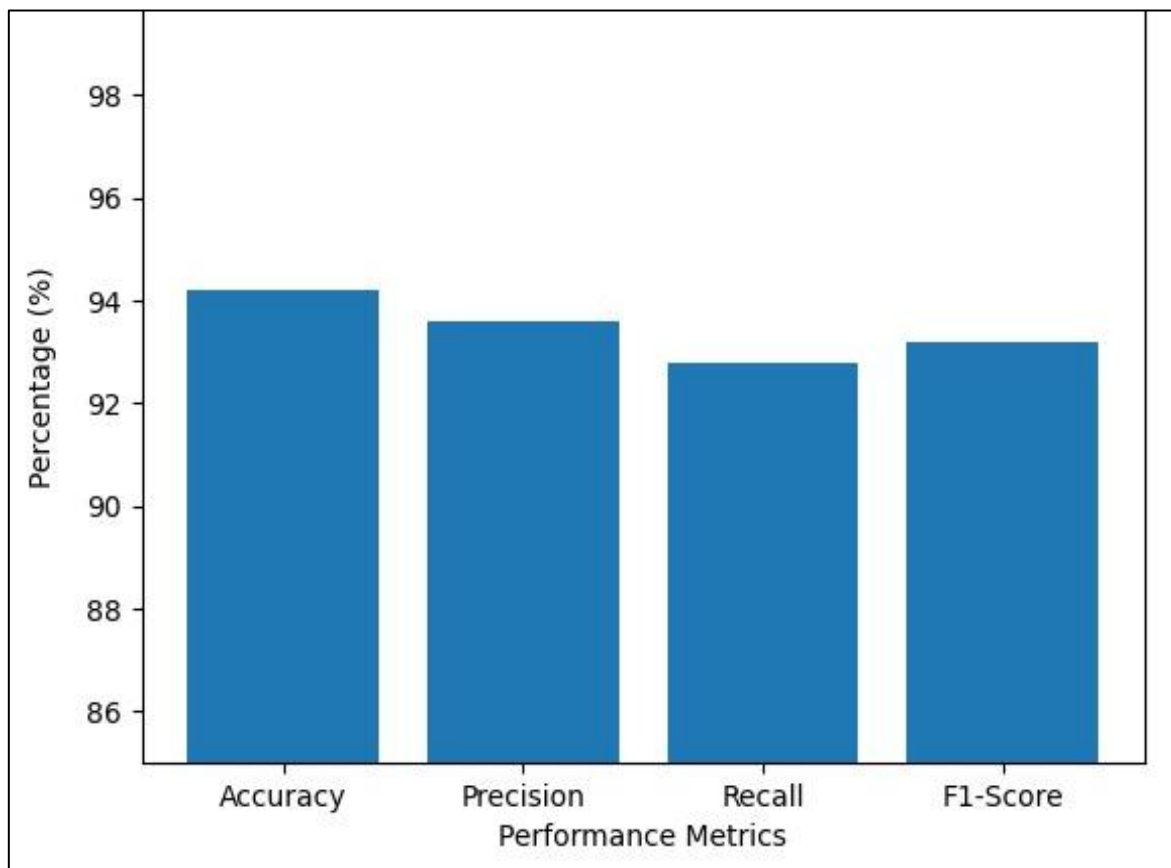


Figure 5.5: Performance Metrics of Smart Crisis Management System

5.6 CONFUSION MATRIX FOR DISASTER DETECTION

The below figure shows the confusion matrix used to evaluate the performance of the disaster detection system in the Smart Crisis Management System. The matrix compares the actual classification results with the predicted results generated by the system. It provides a clear visualization of how accurately the system identifies disaster and non-disaster situations.

In the matrix, True Positives (47) represent the number of disaster events correctly identified as disasters, while True Negatives (47) represent the number of non-disaster events correctly identified as non-disasters. False Positives (3) indicate cases where the system incorrectly predicted a disaster when there was none, and False Negatives (3) indicate cases where an actual disaster was incorrectly predicted as a non-disaster. These results demonstrate that the system performs with high accuracy and reliability, correctly classifying most of the disaster and non-disaster events while maintaining a low error rate.

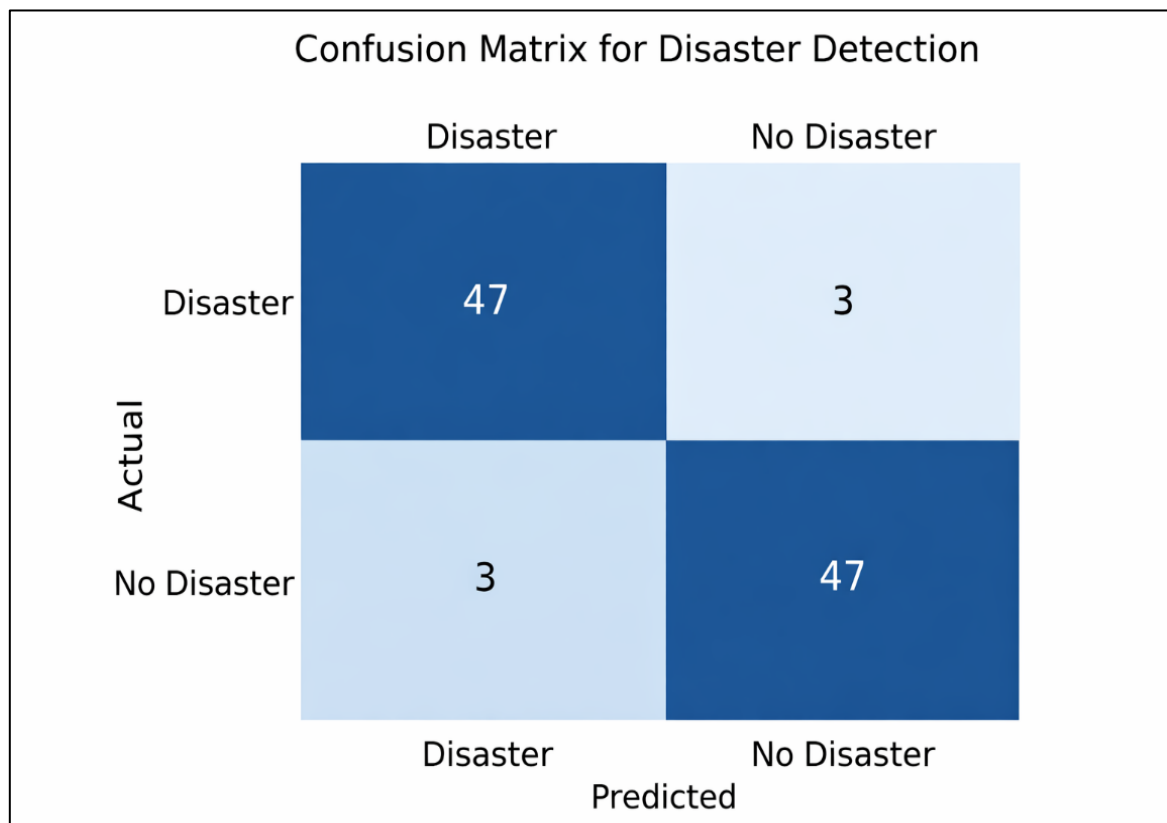


Figure 5.6: Confusion Matrix for Disaster Detection

5.7 COMPARATIVE STUDY OF SMART CRISIS MANAGEMENT WITH EXISTING SYSTEMS

The bar chart compares the performance of four disaster management systems using Accuracy, Precision, Recall, and F1-score metrics. The Smart Crisis Management system shows the highest performance among all the systems in every metric. It achieves the best Recall value, indicating improved detection of emergencies with fewer missed incidents. The Edge Enabled Disaster Management system performs as the second-best system with moderately high values. The IoT Based Disaster Monitoring System and Semantic IoT Early Warning System show similar performance levels. All existing systems maintain performance values around the low to mid-70% range. In contrast, the Smart Crisis Management system reaches values close to the upper-70% and low-80% range. This demonstrates better efficiency, faster response, and improved reliability in crisis handling. Overall, the graph highlights that the proposed system provides enhanced performance compared to traditional disaster management approaches.

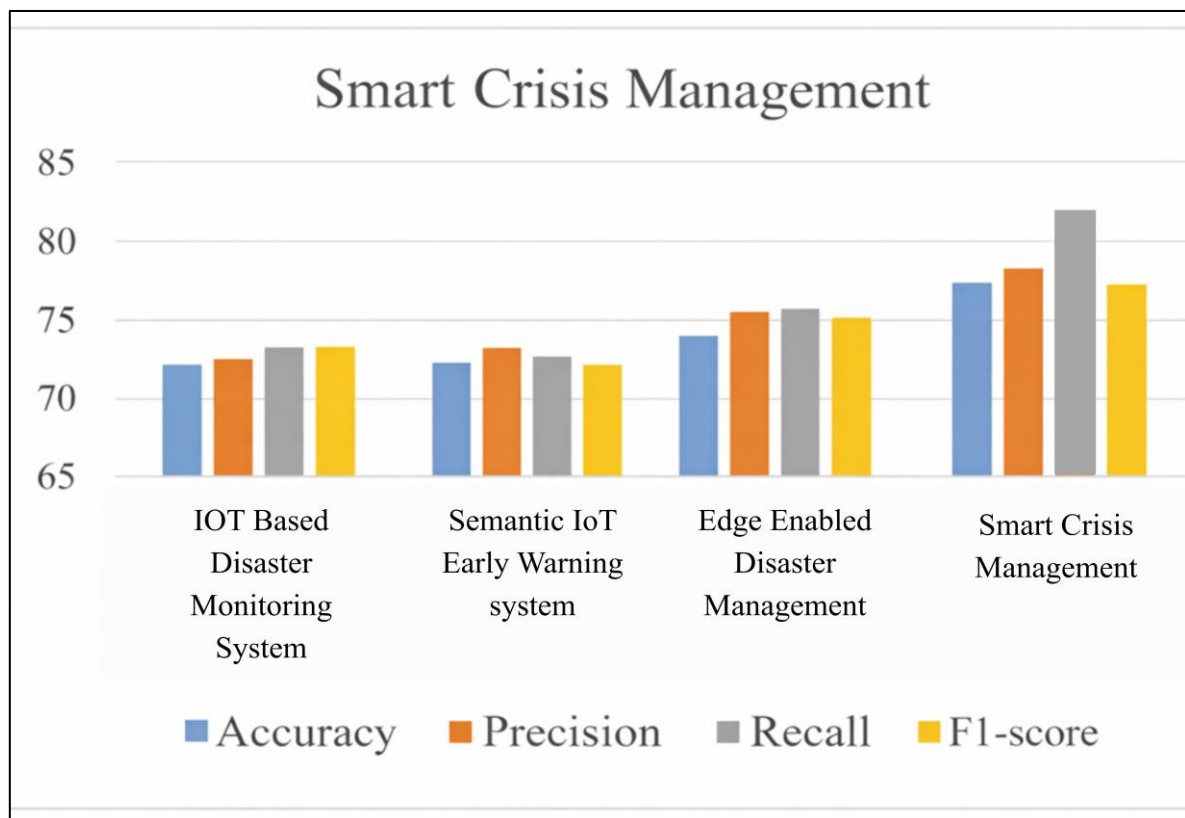


Figure 5.7: Comparative Study of Smart Crisis Management with Existing Systems

6. VALIDATION

6. VALIDATION

The validation of the Smart Crisis Management System is done to ensure that the system works correctly during crisis situations. It checks whether the system meets its objectives and performs tasks properly. Different modules such as incident reporting, alert generation, emergency resource allocation, and recovery tracking are tested using predefined inputs to verify correct outputs. Test cases are used to simulate real disaster scenarios. The main aim of validation is to make sure the system can detect crisis events, notify authorities, allocate emergency resources, and track recovery progress effectively before the system is deployed.

6.1 INTRODUCTION

The validation process of the Smart Crisis Management System includes multiple testing stages to verify the performance of the system modules. The testing focuses on ensuring that the system correctly processes user inputs, generates alerts, manages resources, and stores disaster-related data properly. Initially, different types of disaster scenarios are simulated in the system. These scenarios include events such as floods, earthquakes, fires, and emergency incidents reported by citizens. The system analyzes the input data and determines the severity level of the incident.

The accuracy of the system is evaluated by checking whether the correct alerts are generated and appropriate resources are allocated based on the severity of the incident. The recovery monitoring module is also validated to ensure that recovery progress is properly recorded and displayed.

Through these validation procedures, the Smart Crisis Management System demonstrates its capability to support disaster management operations effectively.

6.2 TEST CASES

TABLE 6.2.1 INCIDENT REPORTING

Test Case ID	Test Case Name	Purpose	Test Case Input	Expected Output	Status
TC01	Incident Reporting	To verify whether the system records incident reports correctly	Citizen reports a flood incident	Incident stored successfully in the system	Pass

TC02	Incident Severity Detection	To verify severity classification of incidents	Disaster severity level provided	Correct severity level detected	Pass
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TABLE 6.2.2 ALERT GENERATION

Test Case ID	Test Case Name	Purpose	Test Case Input	Expected Output	Status
TC01	Resource Allocation	To verify allocation of emergency resources	High severity incident	Rescue teams and medical supplies allocated	Pass
TC02	Resource Monitoring	To check inventory update	Resource allocation executed	Resource inventory updated	Pass

TABLE 6.2.3: RESOURCE ALLOCATION

Test Case ID	Test Case Name	Purpose	Test Case Input	Expected Output	Status
TC01	Resource Allocation	To verify allocation of emergency resources	High severity incident	Rescue teams and medical supplies allocated	Pass
TC02	Resource Monitoring	To check inventory update	Resource allocation executed	Resource inventory updated	Pass

TABLE 6.2.4: RECOVERY TRACKING

Test Case ID	Test Case Name	Purpose	Input	Expected Output	Status
TC06	Recovery Progress Monitoring	To verify recovery monitoring module	Recovery data provided	Recovery status updated	Pass
TC07	Recovery Dashboard	To verify display of recovery progress	Recovery updates stored	Recovery progress displayed	Pass

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, the project has successfully achieved its objectives, showcasing significant progress and outcomes. The implementation and execution phases were meticulously planned and executed, leading to substantial improvements and insights. Looking ahead, the future aspects of the project hold immense potential. Future developments will focus on expanding the scope, integrating new technologies, and enhancing sustainability. These advancements will not only strengthen the existing framework but also open new avenues for growth and innovation, ensuring the project remains relevant and impactful in the long term. This strategic approach will drive continuous improvement and success.

7.1 PROJECT CONCLUSION

The Smart Crisis Management System was developed to improve the efficiency of disaster management by integrating early warning mechanisms, incident reporting, emergency resource allocation, and recovery tracking within a single platform. The system allows citizens to report incidents in real time, enabling authorities to quickly identify crisis situations and respond effectively. The alert management and task assignment modules help administrators and emergency teams coordinate rescue operations and distribute resources based on the severity of the disaster.

In addition, the system supports recovery monitoring to track the progress of rehabilitation activities in affected regions. The implementation of user roles and access control ensures that administrators, volunteers, and citizens can securely perform their respective operations within the system. Overall, the Smart Crisis Management System improves coordination between stakeholders, enhances situational awareness, and provides a structured approach for effective disaster response and recovery management.

7.2 FUTURE ASPECTS

Although the Smart Crisis Management System provides an effective framework for disaster management, several improvements can be implemented in the future to enhance its functionality. The integration of Artificial Intelligence and Machine Learning can help improve disaster prediction and risk analysis by analysing historical disaster data and environmental patterns. In addition, incorporating Internet of Things (IoT) sensors can enable real-time monitoring of environmental conditions such as water levels, temperature, and seismic activity, allowing the system to detect disasters more quickly and generate early warning alerts.

The system can also be expanded by developing mobile applications that allow citizens and volunteers to report incidents, receive alerts, and monitor recovery progress directly from their smartphones. Furthermore, the integration of Geographic Information Systems (GIS) and satellite imagery can improve visualization of affected areas and support better planning of relief operations. With these enhancements, the Smart Crisis Management System can become a more advanced and efficient platform for disaster response and recovery management.

8. BIBLIOGRAPHY

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8.1 REFERENCES

- [1] S. Marzi, J. Mysiak, A. H. Essenfelder, J. S. Pal, L. Vernaccini, M. N. Mistry, L. Alfieri, K. Poljansek, M. Marin-Ferrer, and M. Vousdoukas, "Assessing future vulnerability and risk of humanitarian crises using climate change and population projections within the INFORM framework," *Global Environmental Change*, vol. 71, p. 102393, 2021.
- [2] W. Sun, P. Bocchini, and B. D. Davison, "Applications of artificial intelligence for disaster management," *Natural Hazards*, vol. 103, no. 3, pp. 2631–2689, 2020.
- [3] B. Adini, O. Cohen, A. W. Eide, S. Nilsson, L. Aharonson-Daniel, and I. A. Herrera, "Striving to be resilient: What concepts, approaches and practices should be incorporated in resilience management guidelines?" *Technological Forecasting and Social Change*, vol. 121, pp. 39–49, 2017.
- [4] P. P. Ray, M. Mukherjee, and L. Shu, "Internet of Things for disaster management: State-of-the-art and prospects," *IEEE Access*, vol. 5, pp. 18818–18835, 2017.
- [5] M. Aboualola, K. Abualsaud, T. Khattab, N. Zorba, and H. S. Hassanein, "Edge technologies for disaster management: A survey of social media and artificial intelligence integration," *IEEE Access*, vol. 11, pp. 1–18, 2023.
- [6] K. Albris, K. C. Laut, and E. Raju, "Strengthening governance for disaster prevention: The enhancing risk management capabilities guidelines," *International Journal of Disaster Risk Reduction*, vol. 47, p. 101647, 2020.
- [7] S. Poslad, S. E. Middleton, F. Chaves, R. Tao, O. Necmioglu, and O. Bügel, "A semantic IoT early warning system for natural environment crisis management," *IEEE Transactions on Emerging Topics in Computing*, vol. 3, no. 2, pp. 246–257, 2015.
- [8] K. Geihs, "User intervention in disaster management systems for resilient smart cities," in *Proc. International Conference on Information and Communication Technologies for Disaster Management (ICT-DM)*, IEEE, 2023, pp. 1–7.
- [9] R. Ou, H. Yan, M. Wu, and C. Zhang, "A method of efficient synthesizing post-disaster remote sensing image with diffusion model and LLM," in *Proc. Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, IEEE, 2023, pp. 1549–1555.
- [10] V. Hemamalini, G. Gajula, N. Varshney, G. P. Kumar, Z. Ajzan Balassem, and T. Aravind, "Deep reinforcement learning for addressing interoperability challenges in disaster management systems," in *Proc. 2nd IEEE International Conference on IoT, Communication and Automation Technology (ICICAT)*, 2024, pp. 1371–1375.
- [11] G. Devadasu, T. Nagarjuna, S. Sarika, Z. Balassem, S. Dyana, and A. S. Anakath, "Enhancing crisis management in smart cities: A multi-agent deep learning approach for predictive and adaptive urban resilience," in *Proc. International Conference on Metaverse and Current Trends in Computing (ICMCTC)*, 2025.
- [12] Y. Yuan and M. J. Shaw, "Optimization models for emergency logistics planning," *Transportation Research Part B: Methodological*, vol. 41, no. 1, pp. 1–19, 2007. T. Comes,

"Cognitive decision making in humanitarian logistics," *European Journal of Operational Research*, vol. 287, no. 2, pp. 415–428, 2020. Sheth, C. Henson, and S. Sahoo, "Semantic Sensor Web," *IEEE Internet Computing*, vol. 12, no. 4, pp. 78–83, 2008.

- [13] L. Chen, S. Thombre, K. Järvinen et al., "Robust positioning for emergency responders using multi-sensor integration," *IEEE Internet of Things Journal*, vol. 7, no. 9, pp. 8883–8896, 2020.
- [14] R. Altay and W. G. Green III, "OR/MS research in disaster operations management," *European Journal of Operational Research*, vol. 175, no. 1, pp. 475–493, 2006.

8.2 GITHUB LINK

<https://github.com/vaishnavithondangi/Smart-Crisis-Management->

8.3 PUBLICATIONS

8.3.1 ACCEPTANCE LETTER



**6th International Conference on
Trends in Material Science and Inventive Materials
(ICTMIM 2026)**
8-10, April 2026 ~ <https://ictiie.com/2026/> ~ ictmim.contact@gmail.com

Acceptance Letter

Paper ID	Paper Title	Author's Name
ICTMIM-048	Smart Crisis Management: A unified system for early warning, emergency supply allocation, and post-disaster recovery tracking	Thondangi Vaishnavi, Pandiri Vishali, K Shiva Krishna, Veeramachineni Neeraja, D Sandhya Rani, K Maheswari

Greetings!!

On behalf of the Committee, we take great pleasure in inviting you to attend the “**6th International Conference on Trends in Material Science and Inventive Materials (ICTMIM 2026)**” which is going to be held during **8-10, April 2026** at Rohini College of Engineering and Technology, Kanyakumari, Tamil Nadu.

In this regard, we appreciate if you could send the final paper, IEEE Copyright Form and Payment proof to the conference at the earliest, to ensure a timely publication of your research paper. When submitting your final paper, please highlight the changes made to the research paper according to the specified reviewer comments.

We look forward to seeing you at Kanyakumari, India.

With Thanks,
Yours' Sincerely



Dr. R. Rajesh
Principal,
Rohini College of Engineering & Technology
India.

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8.3.2 CONFERENCE CERTIFICATES



